

PRATIKSHA PENDSE

SOFTWARE ENGINEER – HARDWARE MODELING, SIMULATION & DSP

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Software Engineer with an Electronics (B.E.) and Software Engineering (M.Sc.) background who builds software for hardware development: models, validation tools and the interfaces around them. At Rohde & Schwarz I designed a real-time aliasing detector for oscilloscopes from research to tested FPGA implementation, and built Python tools for IC design and test teams.

SKILLS

Programming	Python, C++, C, MATLAB, VHDL, SQL
Software design	Object-oriented Python, design patterns, Pydantic data models, test-driven development (pytest)
CI/CD & quality	GitLab CI/CD pipelines, tox, mypy, flake8, pylint, code reviews, Git
Modeling & sim	Behavioural modelling in MATLAB, VHDL RTL design, Questa simulation, model-vs-RTL verification
Signal processing	Multirate DSP, decimation, CIC filtering, fixed-point arithmetic
Tools & UI	Streamlit, Plotly, wxPython, NetworkX, Pandas, NumPy, webhooks, Linux, ROS, OpenCV
Languages	English C1, German B2 (in progress)

EDUCATION

M.Sc. Automotive Software Engineering , Chemnitz University of Technology – Grade 2.0	Oct 2022 – Sept 2026
B.E. Electronics and Communication Engineering , KLS Gogte Institute of Technology, India – CGPA 9.11	Aug 2017 – Aug 2021

EXPERIENCE

Master Thesis Student, Signal Processing – Oscilloscopes

Sept 2025 – July 2026

Rohde & Schwarz GmbH & Co. KG, Chemnitz & TU Chemnitz, Chair of Computer Engineering

- Converted a user problem, unreliable decimated waveforms, into a real-time aliasing detector for oscilloscopes.
- Surveyed the state of the art and designed the hardware/software system architecture of the solution.
- Modelled the detector in MATLAB, implemented it in VHDL and verified the two against each other in simulation.
- Analysed accuracy-versus-resource trade-offs and identified possible limitations and directions for future work.

Working Student, Software Tool Development and Testing

Feb 2025 – Aug 2025

Rohde & Schwarz GmbH & Co. KG, Chemnitz

- Built a web dashboard that compares test results across devices and software versions for the development teams.
- Replaced manual report comparison with interactive online plots, so developers see differences at a glance.
- Set up real-time temperature tracking for the test lab, with automatic Microsoft Teams alerts via webhooks.

Intern, Software Tool Development

June 2024 – Nov 2024

Rohde & Schwarz, Dresden

- Co-developed a Python tool that validates ASIC/FPGA system descriptions feeding MATLAB, SystemC and VHDL generators.
- Built layered validation and cross-component checks with object-oriented Pydantic models and NetworkX graphs.
- Delivered test-first through a GitLab CI/CD pipeline with pytest, mypy, flake8 and mandatory code reviews.

Student Assistant, Computer Vision and Robotics

Apr 2023 – Oct 2024

Technische Universität Chemnitz

- Implemented LiDAR-based human tracking in ROS and C++, validated in Gazebo simulation.

Systems Engineer, ITSM Development

Aug 2021 – Jan 2023

Tata Consultancy Services, India

- Developed ITSM workflows, integrations and custom web front-ends, and trained internal teams on adoption.

PUBLICATIONS

- Automotive LiDAR – review.
- Analysis of Automotive Sensor Data: Enhancing Performance and Safety – review.
- Impact of Generative AI Tools on Self-Efficacy of Novice Coders – empirical study.